

```

# This Python 3 environment comes with many helpful analytics
libraries installed
# It is defined by the kaggle/python Docker image:
https://github.com/kaggle/docker-python
# For example, here's several helpful packages to load

import numpy as np # linear algebra
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv)

# Input data files are available in the read-only "../input/"
directory
# For example, running this (by clicking run or pressing Shift+Enter)
will list all files under the input directory

import os
for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))

# You can write up to 20GB to the current directory (/kaggle/working/)
that gets preserved as output when you create a version using "Save &
Run All"
# You can also write temporary files to /kaggle/temp/, but they won't
be saved outside of the current session

/kaggle/input/heights/Heights.csv

```

Reading the data

```

df = pd.read_csv('/kaggle/input/heights/Heights.csv')
df.head()

```

	Weight	Shoesize	Height
0	89.0	10	180.00
1	59.0	7	167.64
2	74.0	10	153.00
3	59.0	7	165.00
4	59.0	100	1650.00

```

X = df.drop(columns=['Height'])
y = df['Height']

```

Statistics of the dataset features

```

X.describe()

```

	Weight	Shoesize
count	37.000000	37.000000
mean	67.609730	10.837838
std	9.769223	15.137359
min	49.000000	5.000000
25%	59.000000	7.000000
50%	70.000000	9.000000
75%	74.000000	10.000000
max	89.000000	100.000000

```
y.describe()
```

count	37.000000
mean	199.146486
std	247.993864
min	5.110000
25%	160.000000
50%	167.000000
75%	175.000000
max	1650.000000

Name: Height, dtype: float64

Filtering the data

```
# First quartile
Q1 = df.quantile(0.25)
# Third quartile
Q3 = df.quantile(0.75)

# Inter quartile range
IQR = Q3 - Q1

mask = ~((df < (df - 1.5 * IQR)) | (df > (Q3 + 1.5 *
IQR))).any(axis=1)

df_clean = df[mask]

X_clean = df_clean.drop(columns=['Height'])
y_clean = df_clean['Height']

X_clean
```

	Weight	Shoesize
0	89.00	10
1	59.00	7
2	74.00	10
3	59.00	7
5	70.00	6
6	73.00	10

7	62.00	7
8	67.00	7
9	61.00	8
10	70.00	9
11	60.00	9
12	84.00	10
13	62.00	8
14	75.00	8
15	75.00	9
16	65.00	8
17	72.00	8
18	51.00	10
19	72.00	10
20	53.00	6
21	57.00	6
22	56.00	7
23	78.00	10
24	49.00	7
25	70.00	10
26	84.00	10
27	65.00	7
28	80.00	8
29	77.56	9
30	55.00	7
31	71.00	9
32	72.00	10
33	58.00	9
34	72.00	10
35	75.00	10
36	70.00	5

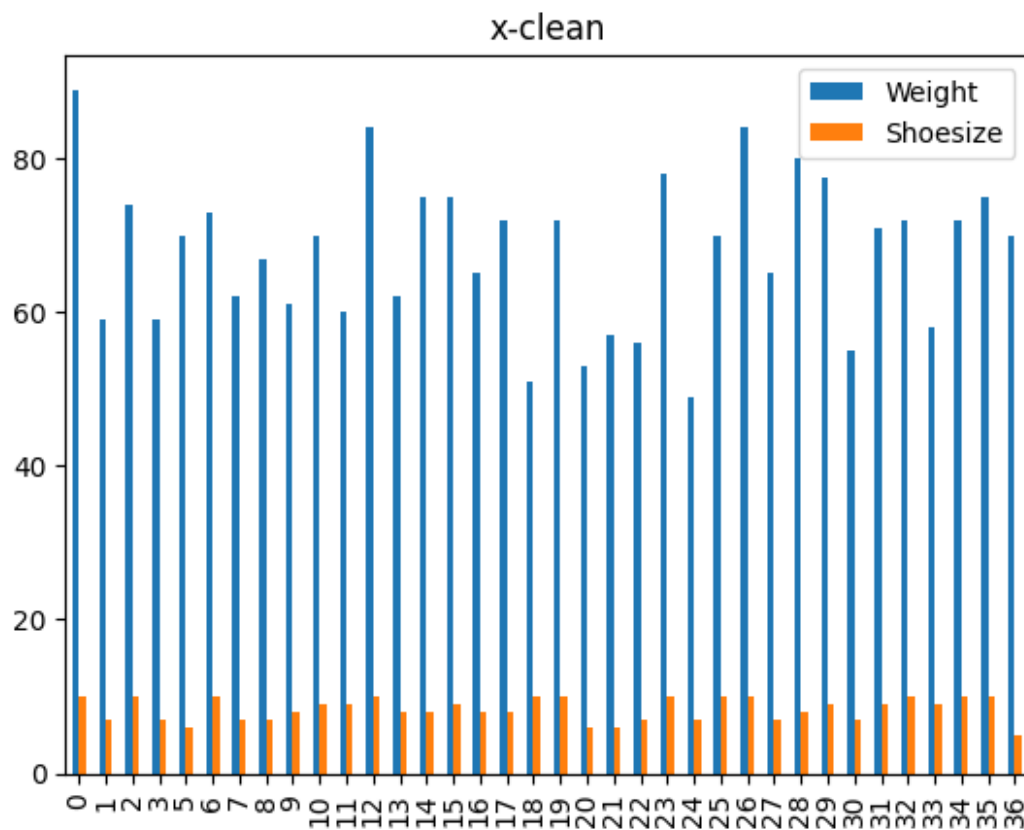
y_clean

0	180.00
1	167.64
2	153.00
3	165.00
5	153.00
6	185.00
7	167.00
8	167.64
9	162.00
10	172.00
11	170.00
12	153.00
13	158.00
14	171.00
15	175.00
16	170.18
17	170.17

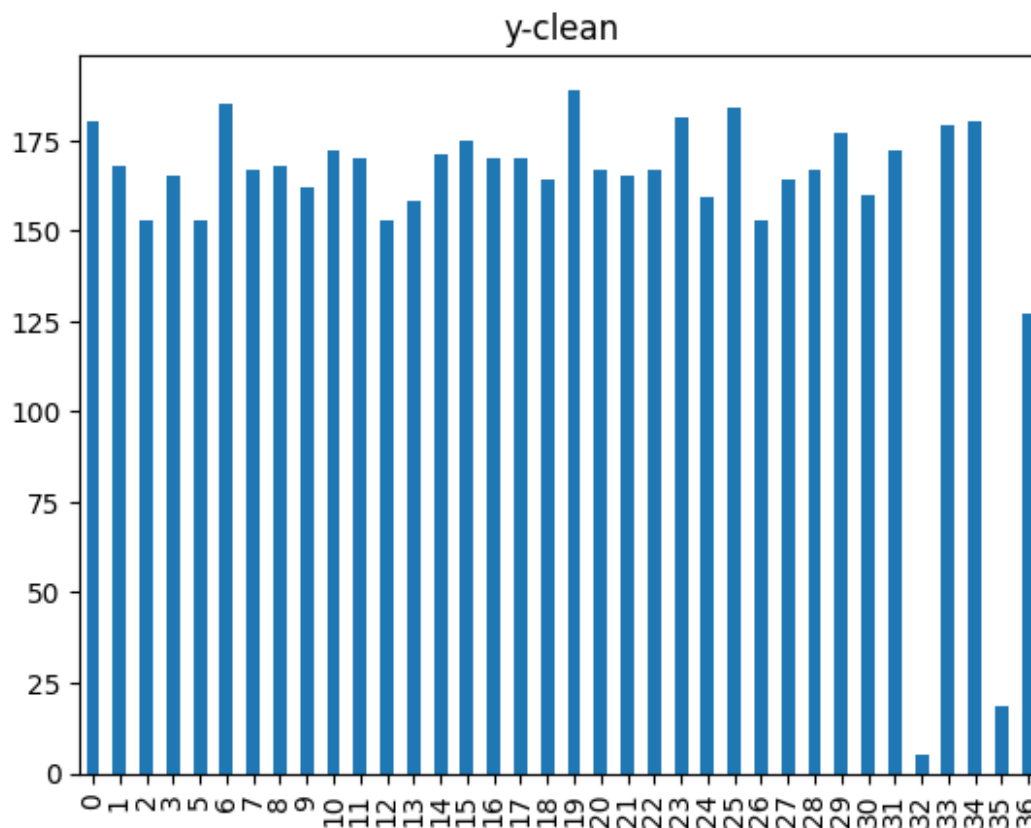
```
18    164.00
19    189.00
20    167.00
21    165.00
22    167.00
23    181.00
24    159.00
25    184.00
26    153.00
27    164.00
28    167.00
29    177.00
30    160.00
31    172.00
32     5.11
33    179.00
34    180.34
35     18.34
36    127.00
Name: Height, dtype: float64
```

Bar plots

```
import matplotlib.pyplot as plt
X_clean.plot(kind='bar',title='x-clean')
<Axes: title={'center': 'x-clean'}>
```



```
y_clean.plot(kind='bar', title='y-clean')  
<Axes: title={'center': 'y-clean'}>
```



Printing shoesize>10 instances

```
df[df['Shoesize']>=10]
```

	Weight	Shoesize	Height
0	89.0	10	180.00
2	74.0	10	153.00
4	59.0	100	1650.00
6	73.0	10	185.00
12	84.0	10	153.00
18	51.0	10	164.00
19	72.0	10	189.00
23	78.0	10	181.00
25	70.0	10	184.00
26	84.0	10	153.00
32	72.0	10	5.11
34	72.0	10	180.34
35	75.0	10	18.34

Custom Dataframe

```
# CUSTOM DATAFRAME
```

```
people_dict = {  
    "weight" : pd.Series([80,67,65],index=['charles','bob','harry']),  
    "height" :  
pd.Series([167,180.33,164],index=['charles','bob','harry']),  
    "father-name" :  
pd.Series(['David','Steve','Pat'],index=['charles','bob','harry'])  
}
```

```
custom_df = pd.DataFrame(people_dict)  
custom_df
```

	weight	height	father-name
charles	80	167.00	David
bob	67	180.33	Steve
harry	65	164.00	Pat